PRACTICAL 11

IMPLEMENTING ADVANCED VISUALIZATION TECHNIQUES AND CUSTOMIZATIONS

Dataset used:

You can use a dataset with fields such as\:You can use your existing dataset too—the field names may be different, but the procedure is similar.

Employee	Department	State	Salary	Year
Amit	IT	Maharashtra	50000	2024
Ravi	IT	Maharashtra	55000	2025
Priya	HR	Gujarat	45000	2024
Neha	HR	Gujarat	52000	2025
Kiran	Sales	Goa	35000	2024
Sneha	Sales	Goa	40000	2025
Rahul	Finance	Rajasthan	48000	2024
Pooja	Finance	Rajasthan	55000	2025
Arjun	IT	Punjab	42000	2024
Meena	IT	Punjab	47000	2025
Sameer	Sales	Karnataka	60000	2024
Anjali	HR	Kerala	50000	2024
Nitin	Sales	Karnataka	68000	2025
Riya	HR	Kerala	58000	2025
Varun	Finance	Tamil Nadu	62000	2024
Kavya	Finance	Tamil Nadu	70000	2025

A) CREATE A SLOPE CHART

What is a slope chart?

A slope chart shows how a value changes between two points, usually two years.

For example:

2024 Salary → 2025 Salary

Step 1: Open Tableau

1. Open Tableau.
2. Open your workbook.
3. Go to a new Worksheet.
4. Rename it: **Slope Chart**

Step 2: Create the basic chart

Drag:

- Department → Rows
- Salary → Columns
- Year → Columns

You may initially get a bar-type visualization.

Step 3: Change Marks type

On the Marks card:

5. Click the drop-down currently showing Automatic.
6. Select Line.

Step 4: Connect the two years

Drag:

Year → Path

on the Marks card.

Now Tableau connects the 2024 and 2025 values.

Step 5: Add labels

Drag:

Salary → Label

on the Marks card.

You now have a basic slope chart.

Expected output

You will see something like:

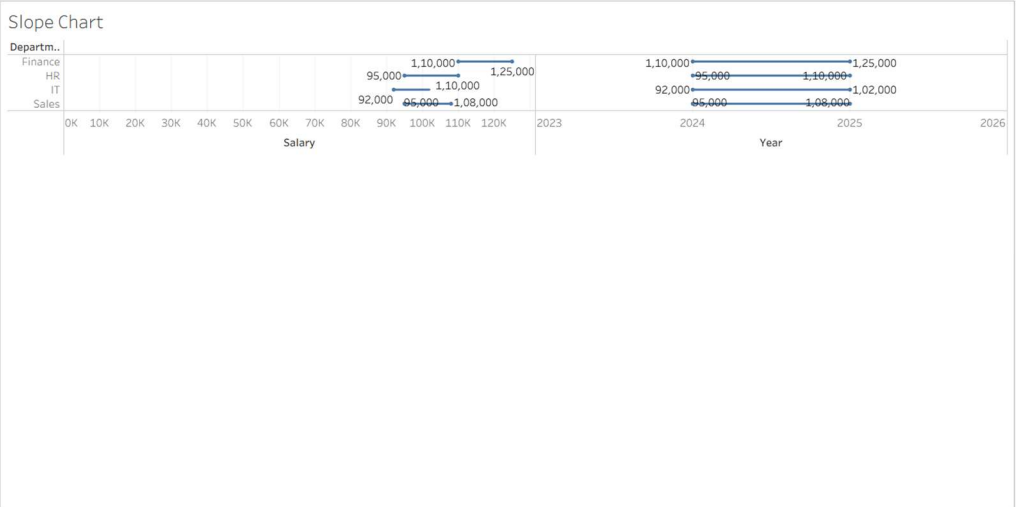
Department

IT ●————————●

HR ●————————●

Sales ●————————————●

The left point represents 2024 and the right point represents 2025.



B) CREATE A LOLLIPOP CHART

What is a lollipop chart?

A lollipop chart is basically a bar chart with a circle at the end.

It makes comparisons visually attractive.

Step 1: Create a new worksheet

Click:

New Worksheet

Rename it:

Lollipop Chart

Step 2: Create the bar

Drag:

Department → Rows

Drag:

Sales → Columns

You will get a horizontal bar chart.

Step 3: Create a dual axis

Drag Sales to the Columns shelf again.

Now you have two Sales axes.

Right-click the second Sales pill.

Select:

Dual Axis

Step 4: Change the first Marks card

You will now see two Marks cards:

- SUM(Sales)
- SUM(Sales) (2)

Select the first one.

Choose:

Bar

Step 5: Change the second Marks card

Select the second Sales Marks card.

Change:

Automatic → Circle

Increase the Size slightly.

Step 6: Synchronize axes

Right-click one of the axes.

Select:

Synchronize Axis

Step 7: Hide the second axis

Right-click the axis.

Select:

Show Header

to remove the duplicate axis if necessary.

Expected result

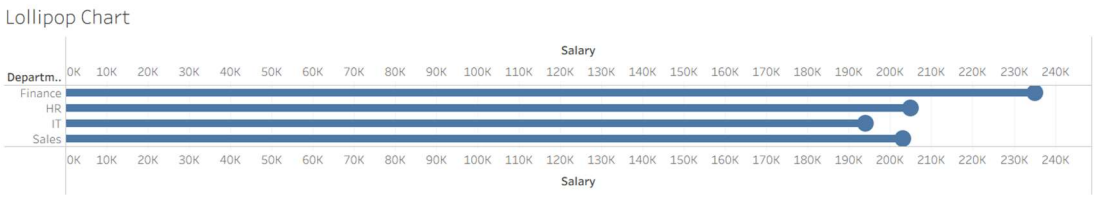
You get:

IT ──────●

HR ─────●

Sales ──────────────●

The bar is the stick and the circle is the lollipop head.



C) DESIGN A WATERFALL CHART

What is a waterfall chart?

A waterfall chart shows how individual values increase or decrease a total.

Example:

Starting Sales → Increase → Decrease → Final Sales

Step 1: Create a new worksheet

Rename:

Waterfall Chart

Step 2: Use a category

Drag:

Department → Columns

Drag:

Sales → Rows

Step 3: Change Marks type

On the Marks card:

Automatic → Gantt Bar

Step 4: Create a calculated field

Go to:

Analysis → Create Calculated Field

Name it:

Sales Base

Formula:

SUM([Sales])

Click OK.

Step 5: Add Sales Base

Drag:

Sales Base → Size

on the Marks card.

You should now see the waterfall effect.

Step 6: Add labels

Drag:

Sales → Label

Step 7: Format

Use:

Color to distinguish increases and decreases if needed.

Expected output

Conceptually:

Starting

[redacted]

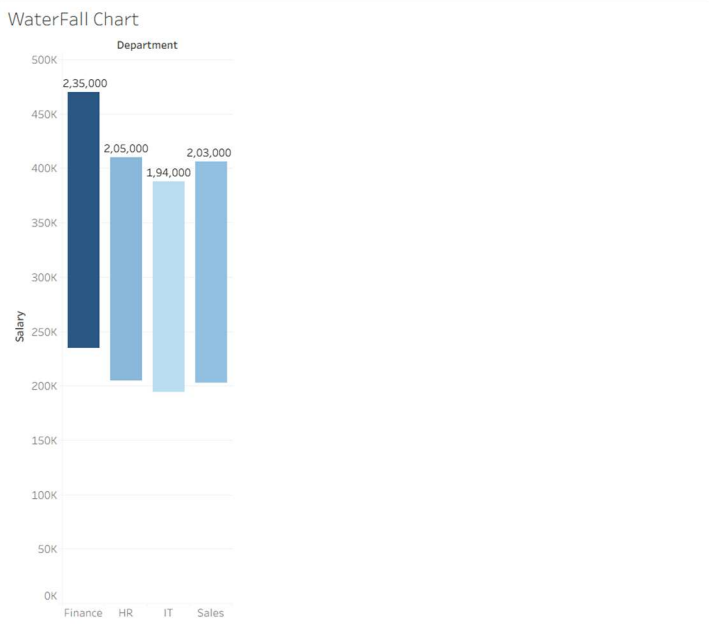
[redacted] Increase

[redacted]

[redacted] Decrease

[redacted] Final

The chart visually explains how different components contribute to the final total.



D) BUILD SPARKLINES FOR TREND ANALYSIS

What is a sparkline?

A sparkline is a small line chart that shows a trend without using a large chart area.

For example:

Department	Trend
IT	
HR	
Sales	

Step 1: Create a new worksheet

Rename:

Sparklines

Step 2: Add Department

Drag:

Department → Rows

Step 3: Add Date/Year

Drag:

Year → Columns

Step 4: Add Sales

Drag:

Sales → Rows

Step 5: Change Marks type

Marks:

Automatic → Line

Now each department has its own small trend line.

Step 6: Remove unnecessary elements

Go to:

Analysis → Table Layout

You can adjust row/column sizing.

You can also hide unnecessary headers/gridlines from:

Format → Lines

Step 7: Make it compact

Reduce the row height so that the chart looks like a small trend indicator.

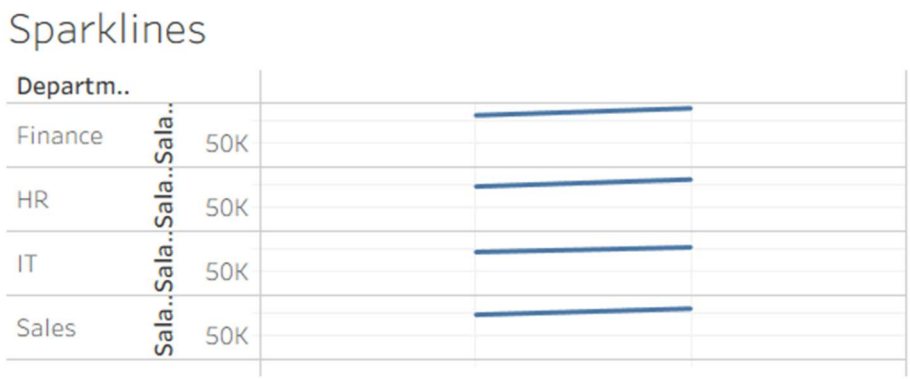
Expected result

IT ／＼＼＿＼＿＼＿／

HR ＼＼＿＼＿／／

Sales ／／／／

The purpose is to quickly identify whether a value is increasing, decreasing, or fluctuating.



E) TRUE SHEET SWAPPING IN TABLEAU

We will swap:

Sales Chart ↔ Salary Chart

using one parameter.

STEP 1: Create the two worksheets

Worksheet 1 — Department Chart

Create:

- Department → Rows
- Employee → Columns

Rename it:

Department Chart

Worksheet 2 — Salary Chart

Create:

- Department → Rows
- Salary → Columns

Rename it:

Salary Chart

STEP 2: Create the parameter

Go to:

Analysis → Create Parameter

Set:

Name: Select Chart

Data type: String

Under List, select:

List of values

Enter:

Department

Salary

Click OK.

STEP 3: Create a calculated field for Department Chart

Go to:

Analysis → Create Calculated Field

Name it:

Show Department

Enter:

[Select Chart] = "Department"

Click OK.

This returns:

- TRUE when Department is selected.
- FALSE when Salary is selected.

STEP 4: Create a calculated field for Salary Chart

Again:

Analysis → Create Calculated Field

Name:

Show Salary

Formula:

[Select Chart] = "Salary"

Click OK.

STEP 5: Put the calculated field on Department Chart

Go to your Department Chart worksheet.

Find:

Show Department

in the Data pane.

Drag:

Show Department → Filters

A Filter window will appear.

Select:

True

Click OK.

Now:

- Select Chart = Department → Department Chart is TRUE → it appears.
- Select Chart = Salary → Department Chart is FALSE → it disappears.

STEP 6: Put the calculated field on Salary Chart

Go to the Salary Chart worksheet.

Drag:

Show Salary → Filters

Select:

True

Click OK.

Now:

- Select Chart = Salary → Salary Chart appears.
- Select Chart = Department → Salary Chart disappears.

STEP 7: Create the Dashboard

Click:

New Dashboard

Drag:

Department Chart

onto the dashboard.

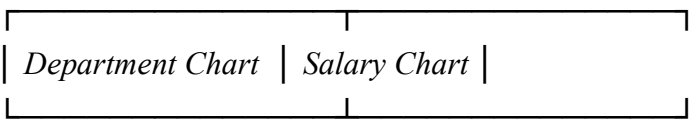
Then drag:

Salary Chart

onto the same location/container.

Important:-

If you simply place them side-by-side:



you will still see two areas.

Instead, put them into the same dashboard area, preferably using a floating layout.

STEP 8: Make the worksheets floating

On the Dashboard:

7. Select Department Chart.
8. Change it to Floating.
9. Select Salary Chart.
10. Change it to Floating.
11. Move them so they occupy exactly the same position and size.

They will overlap.

STEP 9: Show the parameter

On the dashboard, select:

Select Chart

from the worksheet's parameter options and choose:

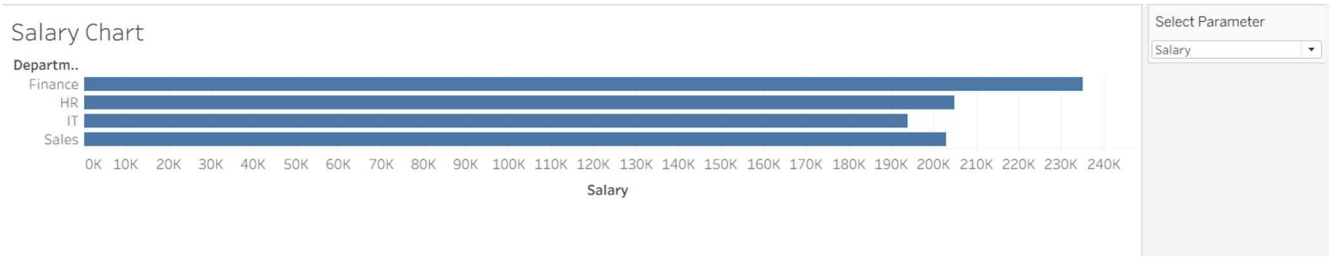
Show Parameter

You should now get a control such as:

Select Chart

Department ▼

Expected Output:-



STEP 10: Test the sheet swap

Choose:

Sales

The dashboard shows:

Department Chart

IT [redacted]

HR [redacted]

Sales [redacted]

Then change the parameter to:

Salary

The Department worksheet becomes FALSE and disappears, while the Salary worksheet becomes TRUE and appears.

Salary Chart

IT [redacted]

HR [redacted]

Sales [redacted]

Then the important part:-

In the Dashboard, click the Sales worksheet:

Department sheet → Filters → Department calculation → select TRUE

Then click the Salary worksheet:

Salary sheet → Filters → Salary calculation → select TRUE

So the result is:

Department selected → Sales Sheet visible

Salary selected → Salary Sheet visible

That is a true sheet swap.

Expected Output:-





F) CREATE CUSTOM GEOGRAPHIC TERRITORIES

What are territories?

Territories allow you to group geographical locations into your own business regions.

For example:

Mumbai

Pune

Nashik

Nagpur

could be grouped as:

Maharashtra Territory

Step 1: Open your geographic worksheet

Create a new worksheet.

Drag:

State → Detail

on the Marks card.

Step 2: Create a group

Right-click:

State

Select:

Create → Group

Step 3: Select states

For example, select:

- Maharashtra
- Gujarat
- Goa

Click:

Group

Rename the group:

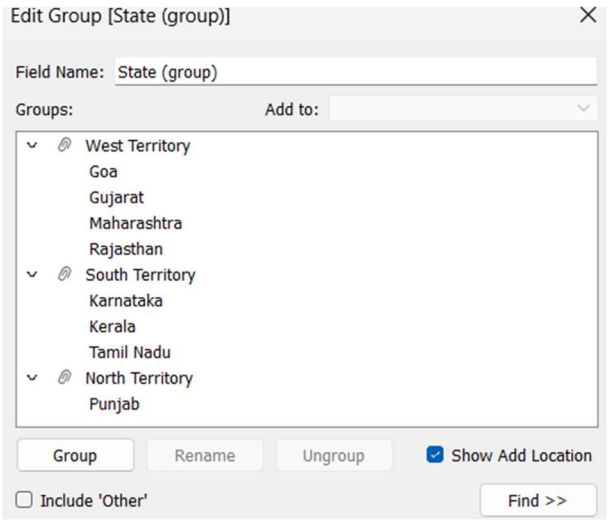
West Territory

Similarly, create:

North Territory

South Territory

etc.



Step 4: Create the map

Double-click:

State

Tableau should automatically create a map if the geographic role is correctly recognized.

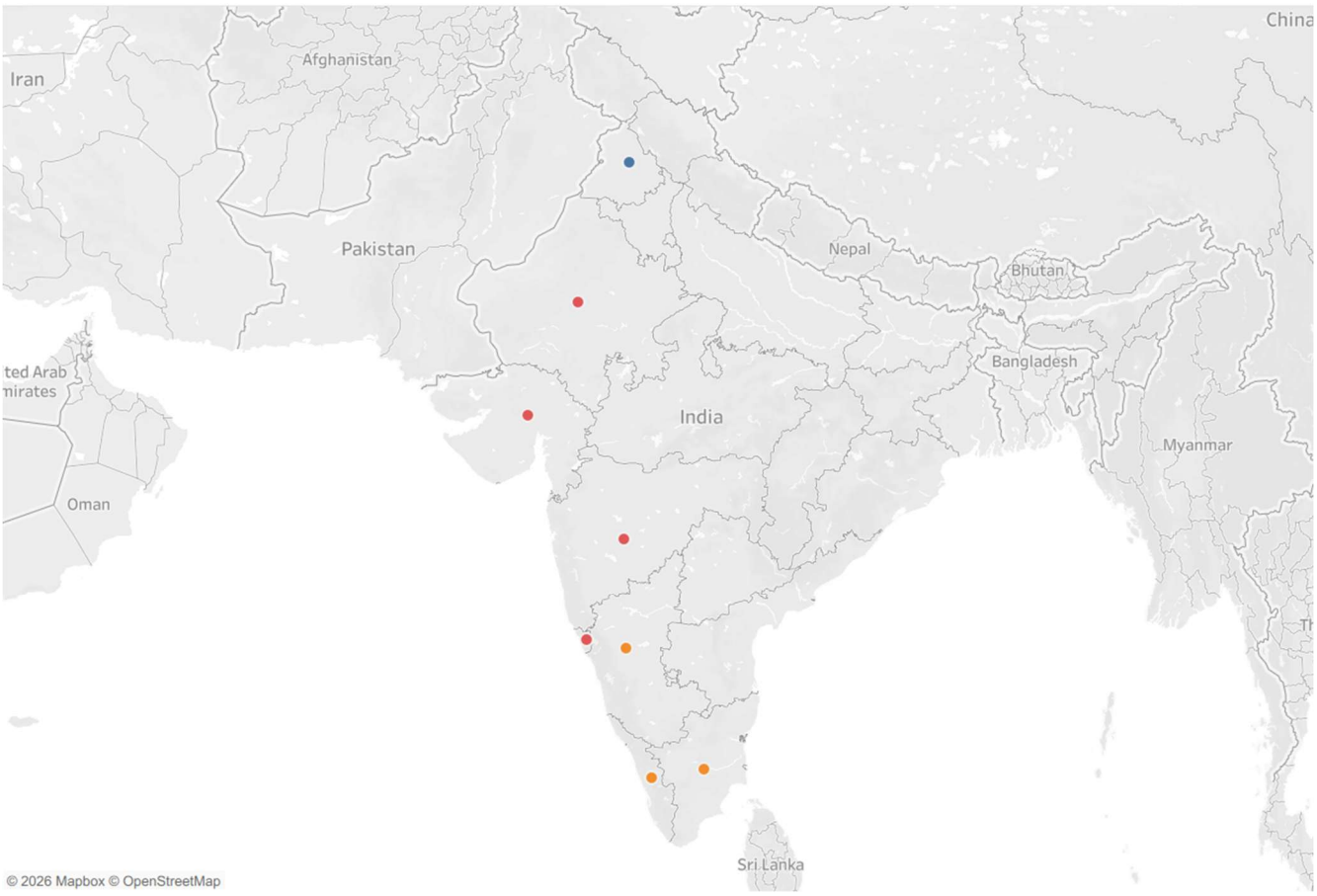
Step 5: Add territory

Drag your newly created:

State Group → Color

Expected output:-

Your map will now show your custom territories, rather than treating every state as a completely separate business region.



If State does not create a map:

Right-click State:

Geographic Role → State/Province

Then double-click it again.

G) USE BACKGROUND IMAGES IN DASHBOARDS

This allows you to place a dashboard visualization over an image.

For example:

Office layout + charts

or

India map image + data visualization

Step 1: Prepare an image

Keep a JPG or PNG image ready.

For example:

office_layout.png

Step 2: Open a dashboard

Click:

New Dashboard

Step 3: Add the image

From the Objects section:

Select:

Image

Step 4: Select your image

Browse to your image.

Click:

Open

Step 5: Adjust the image

Choose:

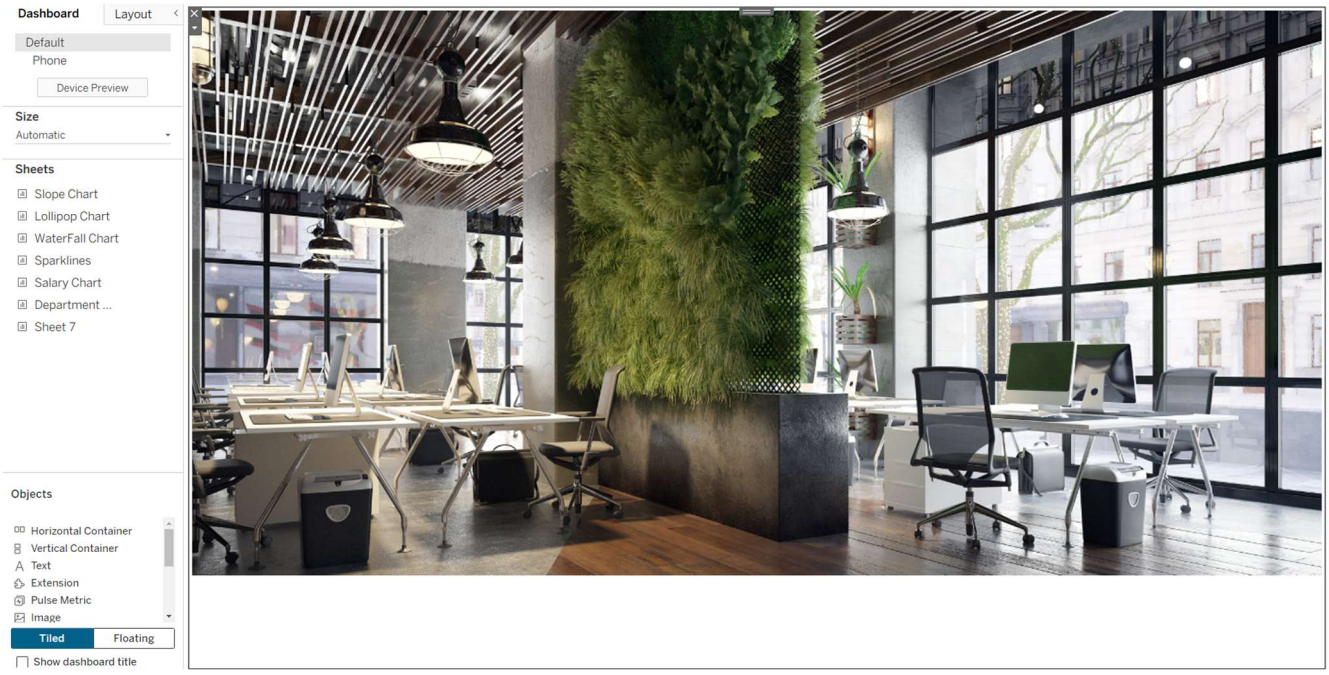
Fit Image

or adjust its size manually.

Important distinction:-

If your practical specifically means a background image behind a worksheet, Tableau also supports adding images through the worksheet/dashboard formatting options depending on the Tableau version.

For a simple practical demonstration, adding an Image object to the dashboard is easiest.



H) APPLY ANIMATION AND TRANSPARENCY

1. Animation

Tableau can animate changes in views when data changes.

Step 1:-

Open your worksheet/dashboard.

Go to:

Format → Animations

In newer Tableau versions, animation controls may appear under:

Format → Animations

Step 2:-

Turn animation:

On

You can adjust:

Duration

For example:

0.5 seconds

Drag Department →Filter

Step 3: Test

Change a filter or parameter.

The chart should transition smoothly instead of changing instantly.

2. Transparency

Transparency controls how see-through a mark is.

Step 1:-

Select the Marks card.

Step 2:-

Click:

Color

Step 3:-

Find:

Opacity

Move the slider.

For example:

100% → completely visible

50% → partially transparent

20% → highly transparent

Why use transparency?

Transparency is useful when many marks overlap.

For example, in a scatter plot:

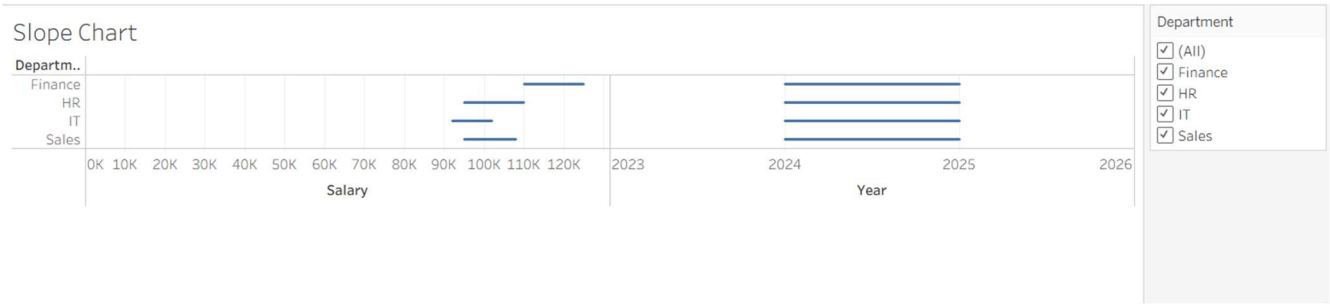
●

●

●●

●●●

Using transparency makes overlapping points easier to identify.



COMPLETE PRACTICAL FLOW

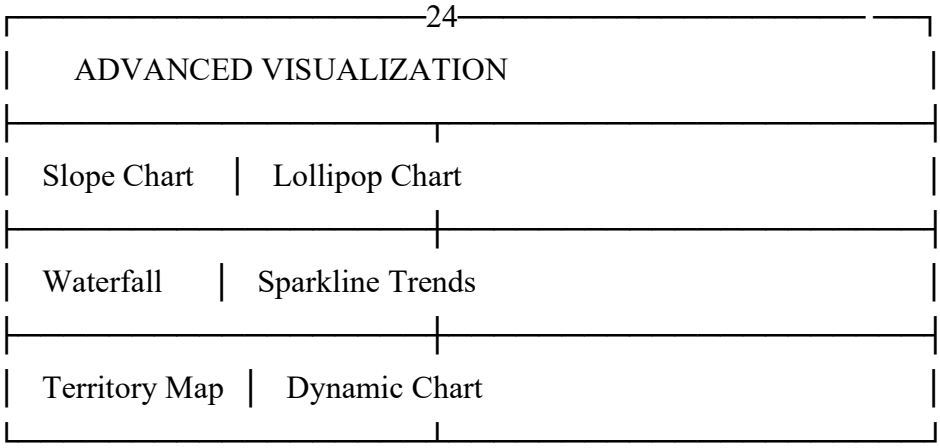
Your Practical 11 can therefore be demonstrated as seven separate worksheets/features:

Part	Visualization/Feature	Main Tableau technique
A	Slope Chart	Line + Path
B	Lollipop Chart	Dual Axis + Bar + Circle
C	Waterfall Chart	Gantt Bar + Calculation
D	Sparkline	Line chart + compact formatting
E	Sheet Swapping	Parameter + Dashboard
F	Custom Territories	Geographic Group
G	Background Image	Dashboard Image
H	Animation & Transparency	Format + Marks → Color

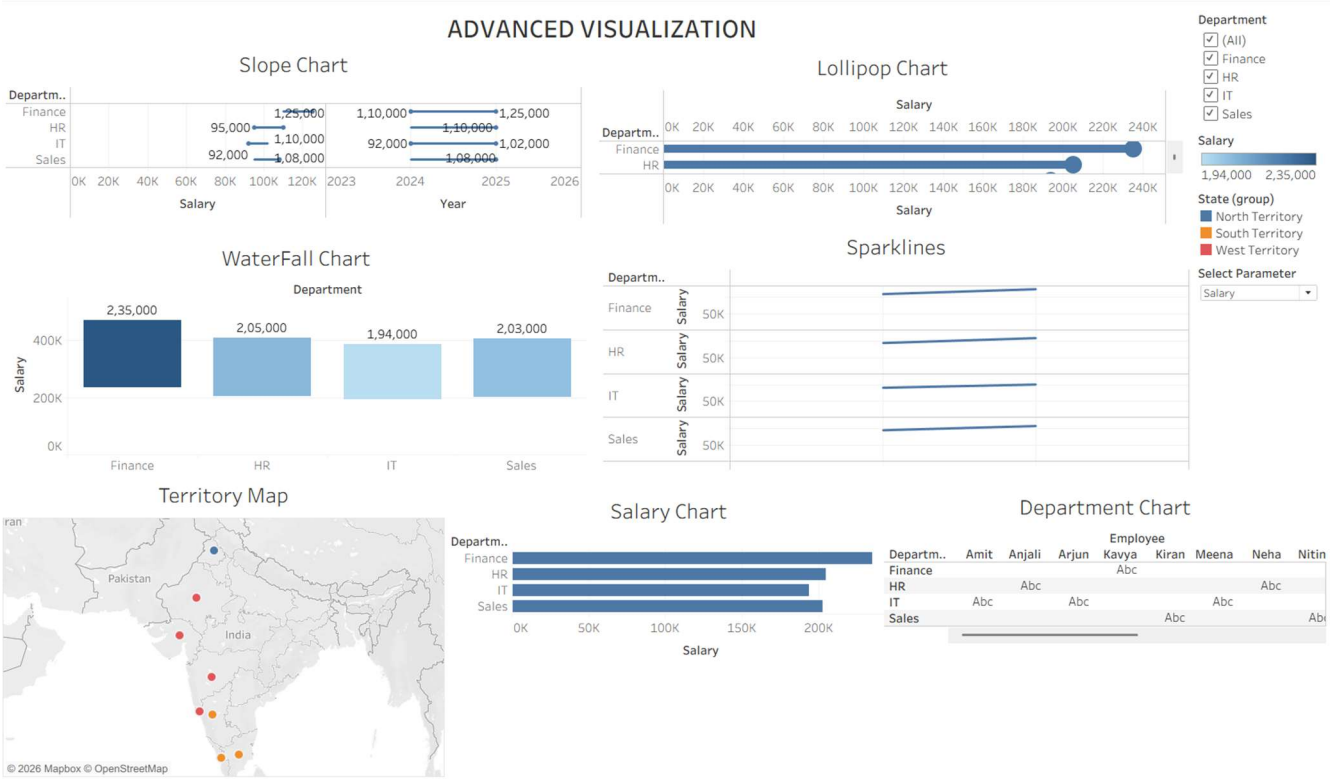
Suggested final dashboard:-

You can create one final dashboard containing:

ADVANCED VISUALIZATION DASHBOARD



This gives you a clear practical demonstration of all the advanced visualization techniques in one Tableau workbook.



Conclusion:-

In this practical, we learned how to create and customize different advanced visualizations and features in Tableau. We explored the creation of slope charts, lollipop charts, waterfall charts, and sparklines to represent data effectively and make comparisons easier. We also learned how to use dual-axis charts, Gantt bars, calculated fields, parameters, and dashboards to enhance data analysis and interactivity. Additionally, we understood how to create custom territories using geographic groups, add background images to dashboards, and apply animation and transparency to improve the visual presentation of data. Through these techniques, we gained practical knowledge of Tableau's advanced charting, formatting, and dashboard customization capabilities. Overall, this practical helped us develop the skills required to transform raw data into meaningful, interactive, and visually appealing representations that support better data interpretation and decision-making.